

Axial vs. Radicular

- Axial
 - Long axis of the body
 - Axial 'views'
- Radicular
 - Nerve root involvement
- 8th Cervical root
 - Levels differ from lumbar.....

Level of disc lesion	C 3/4	L3/4
Central, paracentral, subarticular	C 4 and/or 5 root	L4 root
Foraminal, lateral	C4 root	L3 root

Cervical Roots

- Physical
 - Posture (upper cross pattern)
 - Strength
 - EF (C5), WE (C6), EE (C7), FDP (C8), ADM (T1)
 - Sensation
 - Lateral antecubital (C5), Thumb (C6), Middle finger (C7), Small finger (C8), Medial antecubital (T1)
 - Reflexes
 - Biceps (C5), Brachioradialis (C6), Triceps (C7)

Cervical Roots

- Physical (cont.)
 - Spurling (provocative)
 - Side bend toward, add compression
 - Extend and rotate toward... Side bend toward and rotate toward...
 - Not sensitive; Very specific
 - Abduction (palliative)
 - Decreases tension on cervical roots
 - Also specific but not sensitive

Red Flags

- UMN
 - Pyramidal tracts (Corticospinal)
 - Brain or spinal cord lesion (up to anterior horn)
- Signs
 - Hyperreflexia
 - Clonus & Babinski
 - Spasticity
 - Manual dexterity (direct connection UMN to LMN, no interneurons)
- Gait
 - Spastic (stiff leg, foot drop- circumduction), Falling
- Bowel and Bladder control



cervical



lumbar



cervical



lumbar



cervical



lumbar

Disc

- As a cause for radiculopathy
- Disc as a pain generator itself...
 - Potential (Provocation studies, anatomical studies, biochemical studies, etc.)
 - Degeneration is not pathology (so cant use as proxy for pain with imaging)
- Does it even matter (treatment options)?
 - Anterior Cervical Discectomy & Fusion (ACDF) - no good evidence; 20-30% anecdotal reporting
 - Conclusion: (disc isn't painful or procedure is ineffective)
 - Disc arthroplasty slightly better
 - No interventional procedure

Zygapophyseal Joint

- As a cause for radiculopathy
- As a pain generator itself
 - Degenerative changes
 - Medial branch blocks- diagnostic
 - Radiofrequency ablation- therapeutic
- Which joints refer where...
 - C1-C4: head, neck, shoulder
 - C5-C8: neck, shoulder, arm

Cervicogenic Headache

- C2/3
 - Most prevalent
 - Refers pain: occiput>> parietal/temporal, frontal
- C1/2 and C3/4
 - Second most common
 - Referral patterns vary, but C1/2 tend more superior
- GON and LON
 - Often blocked locally over occiput
 - Limited efficacy

Muscle

- Implicated 20% of chronic neck pain
- Common element in cervicogenic headache

- Postural deviations (including somatic dysfunctions)
- Excess/inefficient usage
- Weak muscles

Shoulder pain

- Common overlap
- Physical exam
 - ROM
 - Hawkins-Kennedy, Neer, Yergason's, Load and Shift
 - Scapular dyskinesia

Instability

- Craniocervical instability
 - EDS/hypermobility
 - Down syndrome (laxity and malformation of dens)
 - Atlantodental Interval (posterior arch of atlas to anterior cortex of dens)
 - X ray (>3mm abnormal)
- Degenerative
 - RA
 - Spondyloarthropathies
 - Joint erosion
 - X ray with flexion/extension (>1-2mm of motion abnormal)

Nerve blocks

- Functional information
- Medial branch (of dorsal ramus of spinal nerve) blocks
 - Diagnostic (Radiofrequency ablation is treatment)
 - C2/3 common for HA
 - C3/4, C4/5 and C5/6 common for neck pain
- Epidural injections
 - Radicular only
 - Higher risk than lumbar

Trauma

- **Whiplash Associated Disorder**
 - Whiplash is the actual mechanism
 - 1.Cervical flexion
 - 2.Lower cervical flexion, upper cervical extension ('S')
 - 3.Lower cervical extension
- **Any structure in the cervical region**
 - Muscle spasm common, loss of lordosis on x ray
 - Majority of patients recover
- **Severity**
 - Grade 1: Subjective symptoms
 - Grade 2: Objective findings (e.g. decreased ROM)
 - Grade 3: Neurological findings (e.g. numbness, decreased reflexes)
 - Grade 4: Fracture